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JEE MAIN PHYSICS DPP: VECTORS RECAP

Topic: Vector Addition, Subtraction, Resolution & Direction Cosines | Time: 45 Min | Max Marks: 80

Marking Scheme: +4, -1 | Designed & Curated by: Syed Mazhar Ali

- Q1.** Two vectors $\vec{A}$ and $\vec{B}$ have equal magnitudes $|\vec{A}| = |\vec{B}| = F$. If the magnitude of their resultant is also equal to F , the angle θ between the two vectors is:
- (a) 60°
(b) 120°
(c) 90°
(d) 150°
- Q2.** The magnitude of the sum of two vectors is equal to the magnitude of their difference, i.e., $|\vec{A} + \vec{B}| = |\vec{A} - \vec{B}|$. The angle between $\vec{A}$ and $\vec{B}$ is:
- (a) 0°
(b) 45°
(c) 90°
(d) 180°
- Q3.** A unit vector in the direction of the resultant of vectors $\vec{A} = 2\hat{i} + 3\hat{j} - \hat{k}$ and $\vec{B} = \hat{i} - \hat{j} + 3\hat{k}$ is:
- (a) $\frac{3\hat{i} + 2\hat{j} + 2\hat{k}}{13}$
(b) $\frac{3\hat{i} + 2\hat{j} + 2\hat{k}}{\sqrt{13}}$
(c) $\frac{3\hat{i} - 2\hat{j} + 2\hat{k}}{\sqrt{17}}$
(d) $\frac{3\hat{i} + 2\hat{j} + 2\hat{k}}{\sqrt{17}}$
- Q4.** A vector $\vec{A}$ makes angles α, β , and γ with the positive x, y , and z axes respectively. The value of $\sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma$ is:
- (a) 2
(b) 1
(c) 3
(d) 0
- Q5.** The resultant of two forces P and Q is R . If force Q is doubled, the new resultant is perpendicular to P . The magnitude of R is equal to:
- (a) P
(b) Q
(c) $\frac{P+Q}{2}$
(d) $\sqrt{P^2 + Q^2}$
- Q6.** Six coplanar forces of equal magnitude 10 N act on a particle. The angle between any two adjacent forces is 60° . The magnitude of their resultant is:
- (a) 60 N
(b) 30 N
(c) 0 N
(d) $10\sqrt{3}$ N
- Q7.** What is the minimum number of coplanar non-zero vectors of unequal magnitudes whose vector sum can be zero?
- (a) 2
(b) 4
(c) 5
(d) 3
- Q8.** What is the minimum number of non-coplanar non-zero vectors whose resultant can be zero?
- (a) 4
(b) 3
(c) 2
(d) 5
- Q9.** A vector $\vec{A} = 3\hat{i} + 4\hat{j}$ is rotated through an angle of 90° in the counter-clockwise direction in the xy -plane. The new vector $\vec{A}'$ is:
- (a) $4\hat{i} - 3\hat{j}$
(b) $-4\hat{i} + 3\hat{j}$
(c) $-3\hat{i} - 4\hat{j}$
(d) $3\hat{i} - 4\hat{j}$
- Q10.** If $\vec{A} + \vec{B} = \vec{C}$ and $|\vec{A}| + |\vec{B}| = |\vec{C}|$, the angle between $\vec{A}$ and $\vec{B}$ is:
- (a) 180°
(b) 90°
(c) 0°
(d) 45°
- Q11.** If $\vec{A} + \vec{B} = \vec{C}$ and $A^2 + B^2 = C^2$, the angle between $\vec{A}$ and $\vec{B}$ is:
- (a) 0°
(b) 45°
(c) 180°
(d) 90°
- Q12.** A displacement vector of magnitude 10 m is inclined at 30° to the x -axis and 60° to the y -axis. Its rectangular components along the x and y axes are:
- (a) $5\sqrt{3}$ m, 5 m
(b) 5 m, $5\sqrt{3}$ m

- (c) 10 m, 5 m
(d) $5\sqrt{2}$ m, $5\sqrt{2}$ m
- Q13.** The maximum and minimum magnitudes of the resultant of two given vectors are 17 units and 7 units respectively. If these two vectors are placed at right angles to each other, the magnitude of their resultant is:
- (a) 12 units
(b) 13 units
(c) 15 units
(d) $\sqrt{119}$ units
- Q14.** Three vectors $\vec{P}$, $\vec{Q}$, and $\vec{R}$ satisfy $\vec{P} + \vec{Q} + \vec{R} = 0$. If $|\vec{P}| = 12$, $|\vec{Q}| = 5$, and $|\vec{R}| = 13$, the angle between $\vec{P}$ and $\vec{Q}$ is:
- (a) 0°
(b) 45°
(c) 90°
(d) 180°
- Q15.** A particle is displaced from position $\vec{r}_1 = (2\hat{i} + 3\hat{j} - \hat{k})$ m to $\vec{r}_2 = (4\hat{i} + 7\hat{j} + 2\hat{k})$ m under the action of a constant force. The magnitude of the displacement vector is:
- (a) $\sqrt{20}$ m
(b) $\sqrt{24}$ m
(c) 7 m
(d) $\sqrt{29}$ m
- Q16.** Given two vectors $\vec{A} = 2\hat{i} + \hat{j}$ and $\vec{B} = \hat{i} - \hat{j}$. The component of vector $\vec{A}$ along the direction of vector $\vec{B}$ is:
- (a) $\frac{1}{\sqrt{2}}$
(b) $\frac{1}{2}$
(c) $\sqrt{2}$
(d) $\frac{3}{\sqrt{2}}$
- Q17.** Which of the following sets of forces acting simultaneously on a particle **cannot** produce equilibrium?
- (a) 3 N, 4 N, 5 N
(b) 2 N, 3 N, 6 N
(c) 5 N, 5 N, 5 N
(d) 10 N, 10 N, 15 N
- Q18.** The angle θ that the vector $\vec{A} = \hat{i} + \hat{j} + \sqrt{2}\hat{k}$ makes with the positive z -axis is:
- (a) 30°
(b) 60°
(c) 45°
(d) 90°
- Q19.** Two vectors $\vec{A}$ and $\vec{B}$ are inclined at an angle θ . If $|\vec{A} + \vec{B}| = \sqrt{3}|\vec{A} - \vec{B}|$ and $|\vec{A}| = |\vec{B}|$, the angle θ is:
- (a) 30°
(b) 45°
(c) 90°
(d) 60°
- Q20.** A force of 10 N is resolved into two mutually perpendicular components. If one of the components is 6 N, the other component is:
- (a) 8 N
(b) 4 N
(c) $\sqrt{136}$ N
(d) 6 N

ANSWER KEY & STEP-BY-STEP SOLUTIONS

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Quick Answer Key

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
1	(b)	6	(c)	11	(d)	16	(a)
2	(c)	7	(d)	12	(a)	17	(b)
3	(d)	8	(a)	13	(b)	18	(c)
4	(a)	9	(b)	14	(c)	19	(d)
5	(b)	10	(c)	15	(d)	20	(a)

Detailed Solutions

Sol 1. Correct Option: (b)

Using parallelogram law:

$$R^2 = A^2 + B^2 + 2AB \cos \theta$$

Given $R = A = B = F$:

$$F^2 = 2F^2(1 + \cos \theta)$$

$$1 + \cos \theta = \frac{1}{2} \implies \cos \theta = -\frac{1}{2} \implies \theta = 120^\circ$$

Sol 2. Correct Option: (c)

$$|\vec{A} + \vec{B}|^2 = |\vec{A} - \vec{B}|^2$$

$$A^2 + B^2 + 2AB \cos \theta = A^2 + B^2 - 2AB \cos \theta$$

$$4AB \cos \theta = 0 \implies \cos \theta = 0 \implies \theta = 90^\circ$$

Sol 3. Correct Option: (d)

$$\vec{R} = \vec{A} + \vec{B} = (2+1)\hat{i} + (3-1)\hat{j} + (-1+3)\hat{k} = 3\hat{i} + 2\hat{j} + 2\hat{k}$$

Magnitude:

$$|\vec{R}| = \sqrt{3^2 + 2^2 + 2^2} = \sqrt{9+4+4} = \sqrt{17}$$

Unit vector:

$$\hat{R} = \frac{\vec{R}}{|\vec{R}|} = \frac{3\hat{i} + 2\hat{j} + 2\hat{k}}{\sqrt{17}}$$

Sol 4. Correct Option: (a)

Direction cosines satisfy $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$.

Using $\sin^2 \theta = 1 - \cos^2 \theta$:

$$\sum \sin^2 \alpha = \sum (1 - \cos^2 \alpha)$$

$$= 3 - \sum \cos^2 \alpha = 3 - 1 = 2$$

Sol 5. Correct Option: (b)

When Q is doubled, resultant $\vec{R}'$ is perpendicular to $\vec{P}$:

$$\tan(90^\circ) \implies P + 2Q \cos \theta = 0 \implies \cos \theta = -\frac{P}{2Q}$$

Resultant R :

$$R^2 = P^2 + Q^2 + 2PQ \left(-\frac{P}{2Q}\right) = Q^2 \implies R = Q$$

Sol 6. Correct Option: (c)

The forces form a closed regular symmetric hexagon because their vector sum $\sum_{k=1}^6 \vec{F}_k = 0$. Three opposite pairs of equal and opposite forces cancel each other completely, giving a net resultant of 0 N.

Sol 7. Correct Option: (d)

Two unequal vectors cannot cancel each other because $|\vec{A}| \neq |\vec{B}|$. Three coplanar vectors of unequal magnitudes can form a closed triangle ($\vec{A} + \vec{B} + \vec{C} = 0$), so the minimum number is 3.

Sol 8. Correct Option: (a)

Three non-coplanar vectors cannot form a closed loop because any two lie in one plane and their resultant lies in that same plane, while the third vector has a component perpendicular to that plane. Hence, a minimum of 4 non-coplanar vectors are required to form a closed polygon in 3D.

Sol 9. Correct Option: (b)

Rotating a 2D vector (x, y) counter-clockwise by 90° maps it to $(-y, x)$:

$$\text{Here } \vec{A} = 3\hat{i} + 4\hat{j} \implies \vec{A}' = -4\hat{i} + 3\hat{j}$$

$$\text{Dot product check: } \vec{A} \cdot \vec{A}' = 3(-4) + 4(3) = 0$$

Sol 10. Correct Option: (c)

$$|\vec{A} + \vec{B}| = A + B$$

$$\sqrt{A^2 + B^2 + 2AB \cos \theta} = A + B$$

Squaring gives:

$$2AB \cos \theta = 2AB \implies \cos \theta = 1 \implies \theta = 0^\circ$$

Sol 11. Correct Option: (d)

$$C^2 = A^2 + B^2 + 2AB \cos \theta$$

$$\text{Given } C^2 = A^2 + B^2:$$

$$2AB \cos \theta = 0 \implies \theta = 90^\circ$$

Sol 12. Correct Option: (a)

$$A_x = A \cos(30^\circ) = 10 \left(\frac{\sqrt{3}}{2}\right) = 5\sqrt{3} \text{ m.}$$

$$A_y = A \sin(30^\circ) = 10 \left(\frac{1}{2}\right) = 5 \text{ m.}$$

Sol 13. Correct Option: (b)

$$R_{\max} = A + B = 17, \text{ and } R_{\min} = A - B = 7.$$

$$\text{Adding: } 2A = 24 \implies A = 12 \text{ units, } B = 5 \text{ units.}$$

When at right angles ($\theta = 90^\circ$):

$$R = \sqrt{A^2 + B^2} = \sqrt{12^2 + 5^2} = 13 \text{ units}$$

Sol 14. Correct Option: (c)

$$\vec{P} + \vec{Q} = -\vec{R} \implies |\vec{P} + \vec{Q}|^2 = |\vec{R}|^2.$$

$$P^2 + Q^2 + 2PQ \cos \theta = R^2$$

$$12^2 + 5^2 + 2(12)(5) \cos \theta = 13^2$$

$$169 + 120 \cos \theta = 169 \implies \theta = 90^\circ$$

Sol 15. Correct Option: (d)

$$\Delta \vec{r} = \vec{r}_2 - \vec{r}_1 = (4-2)\hat{i} + (7-3)\hat{j} + (2-(-1))\hat{k} = 2\hat{i} + 4\hat{j} + 3\hat{k}.$$

Magnitude:

$$|\Delta \vec{r}| = \sqrt{2^2 + 4^2 + 3^2} = \sqrt{4 + 16 + 9} = \sqrt{29} \text{ m}$$

Sol 16. Correct Option: (a)

The scalar component of $\vec{A}$ along $\vec{B}$ is given by:

$$A_{\parallel} = \frac{\vec{A} \cdot \vec{B}}{|\vec{B}|}$$

$$\vec{A} \cdot \vec{B} = (2)(1) + (1)(-1) = 2 - 1 = 1.$$

$$|\vec{B}| = \sqrt{1^2 + (-1)^2} = \sqrt{2}.$$

$$A_{\parallel} = \frac{1}{\sqrt{2}}$$

Sol 17. Correct Option: (b)

For three forces to be in equilibrium, the sum of any two magnitudes must be greater than or equal to the third magnitude ($A + B \geq C$).

For (2 N, 3 N, 6 N):

$$2 + 3 = 5 < 6$$

Hence, this set cannot form a closed triangle and cannot be in equilibrium.

Sol 18. Correct Option: (c)

$$|\vec{A}| = \sqrt{1^2 + 1^2 + (\sqrt{2})^2} = \sqrt{1 + 1 + 2} = \sqrt{4} = 2.$$

Direction cosine with z -axis:

$$\cos \gamma = \frac{A_z}{|\vec{A}|} = \frac{\sqrt{2}}{2} = \frac{1}{\sqrt{2}} \implies \gamma = 45^\circ$$

Sol 19. Correct Option: (d)

$$\text{Given } |\vec{A} + \vec{B}|^2 = 3|\vec{A} - \vec{B}|^2 \text{ with } A = B:$$

$$2A^2(1 + \cos \theta) = 3(2A^2)(1 - \cos \theta)$$

$$1 + \cos \theta = 3 - 3 \cos \theta$$

$$\cos \theta = \frac{1}{2} \implies \theta = 60^\circ$$

Sol 20. Correct Option: (a)

For two perpendicular components F_x and F_y :

$$F^2 = F_x^2 + F_y^2$$

$$10^2 = 6^2 + F_y^2 \implies F_y = \sqrt{100 - 36} = 8 \text{ N}$$

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